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The Arctic Boreal Burned Area (ABBA) Product

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ABSTRACT

The Arctic Boreal Burned Area (ABBA) product is a region-specific burned area product designed for high northern latitudes (HNL), including boreal forests and Arctic tundra. Developed to address limitations of global products like MODIS MCD64A1, ABBA uses MODIS data to calculate the differenced Normalised Burn Ratio (dNBR) and incorporates vegetation cover thresholds to identify burned areas. It includes a spring composite modification for high-latitude tiles to capture late-season fires obscured by snow. ABBA achieves higher accuracy (87.2%) compared to MCD64A1 (82.7%), with lower commission and omission errors, enabling more reliable detection of burned areas. Between 2002 and 2022, ABBA mapped over 2 million km^2 of burned area, significantly more than MCD64A1. Its applications include improving wildfire emission estimates, studying climate feedbacks like albedo changes and assessing ecological impacts such as permafrost thaw and vegetation regrowth. ABBA is available via the Oak Ridge National Laboratory Distributed Active Archive Center (ORNL DAAC).

Dataset Information

Property	Details
Identifier	https://doi.org/10.3334/ORNLDAAC/2328
Creator	Dong Chen et al.
Dataset Correspondence	itscd@umd.edu
Title	Arctic Boreal Annual Burned Area (ABBA), Circumpolar Boreal Forest and Tundra V2
Publisher	Oak Ridge National Laboratory Distributed Active Archive Centre (ORNL DAAC)
Publication Year	2024
Version	V2

1 | Introduction

The high northern latitude (HNL) region, characterised by low annual temperatures, long snow seasons and relatively sparse human presence, encompasses two primary terrestrial biomes: boreal forests and Arctic tundra. In recent decades, this region has experienced some of the most accelerated warming on the planet, with temperatures rising at multiple times the global average rate (Post et al. 2019; Rantanen et al. 2022). This warming has led to significant changes in various natural processes, including fire regimes (Kasischke and Turetsky 2006). With fewer human activities compared to temperate and tropical regions, wildfires represent the dominant disturbance factor in the circumpolar region, exerting profound climatic, ecological and societal impacts (Flannigan et al. 2009). These fires alter carbon cycling by releasing large amounts of stored carbon into the atmosphere (Turetsky et al. 2011), degrade air quality (Chen et al. 2023) and threaten the stability of local communities (Trainor et al. 2009). Given the vastness and remoteness of these northern landscapes, traditional

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ground-based monitoring is often impractical, making satellite remote sensing an indispensable tool (French et al. 2008). Satellite observations enable researchers to capture the spatial and temporal distribution of wildfires across vast areas, track their progression throughout the fire season and analyse long-term impacts such as shifts in vegetation cover and permafrost stability. Consequently, a detailed understanding of wildfire patterns in HNL ecosystems, supported by robust remote sensing methods, is crucial for improving climate models and guiding policy decisions aimed at mitigating adverse effects (McGuire et al. 2018).

Several global burned area products, including MODIS MCD64A1 (Giglio et al. 2018) and FireCCI51 (Lizundia-Loiola et al. 2020), provide coverage of the HNL region. While these products have significantly advanced our understanding of northern wildfire regimes, their performance in high-latitude environments is often suboptimal (Chen, Shevade, et al. 2021; Loboda et al. 2007). This limitation primarily arises from the inability of global algorithms to account for the environmental conditions that are unique to the high latitude regions. Key challenges include long snow seasons and low solar angles, which limit the availability of snow-free imagery for accurate burned area detection and late-season fires, which are often obscured by snow before satellites can capture snow-free images, leading to incomplete mapping of burned areas (Chen, Shevade, et al. 2021). These challenges highlight the need for region-specific approaches to improve the accuracy of burned area mapping in the HNL. To address these limitations, the Arctic Boreal Burned Area (ABBA; Loboda et al. 2024) product was developed as a region-specific alternative to global burned area products. ABBA is designed to account for the unique environmental conditions of the HNL, ensuring more accurate and reliable mapping of wildfires in boreal forests and Arctic tundra ecosystems.

TABLE 1 | Basic parameters of ABBA V2.

Spatial coverage	Circumpolar boreal forests and tundra (MODIS vertical tiles V1–V4)
Temporal domain	2002–2022
Number of files	21
File format	GeoTIFF
Total file size	347 MB
Time step	Annual
Spatial resolution	463.313 m
Projection	Sinusoidal
Burned pixel value	1

1.1 | Product Characteristics

The ABBA product presented here represents its second version (V2). (Table 1) provides a summary of the basic parameters of ABBA, including spatial resolution and temporal coverage. (Figure 1) illustrates the cumulative burned areas mapped by ABBA across the circumpolar boreal forests and tundra regions between 2002 and 2022.

2 | Methods

The ABBA product is developed based on a modified and expanded version of the algorithm proposed by (Loboda

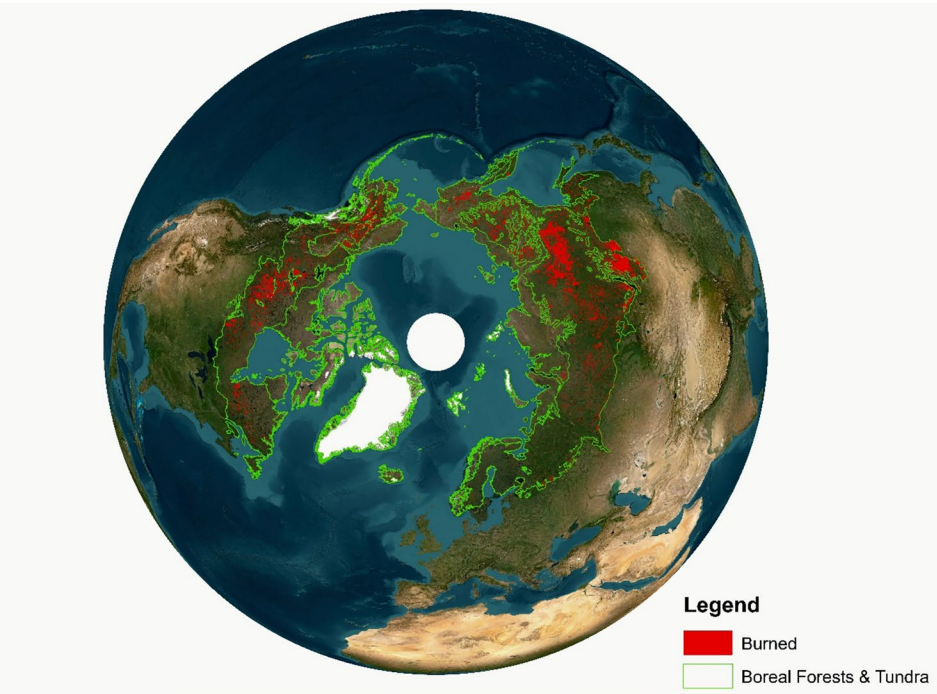

FIGURE 1 | The distribution of cumulative burned area between 2002 and 2022 as mapped by ABBA V2.

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et al. 2007). The following paragraphs summarise the key steps in the ABBA production workflow. The algorithm has an additional spring composite modification for MODIS Vertical Tiles #1 (V1) and Vertical Tiles #2 (V2) compared to Vertical Tiles #3 (V3) and Vertical Tiles #4 (V4).

The production of ABBA began with the creation of the differenced Normalised Burn Ratio (dNBR; Key and Benson 2006). This involved calculating the Normalised Burn Ratio (NBR; Garcia and Caselles 1991) using MODIS MOD09A1 surface reflectance data (Collection 6.1; Vermote 2021), specifically bands 2 (near infrared; NIR) and 7 (shortwave infrared at 2.1 μm wavelength; SWIR2.1), following the formula:

$$\mathrm{NBR} = \left(\mathrm{SR}_{\mathrm{NIR}} - \mathrm{SR}_{\mathrm{SWIR2.1}}\right) / \left(\mathrm{SR}_{\mathrm{NIR}} + \mathrm{SR}_{\mathrm{SWIR2.1}}\right)$$

where $\mathrm{SR}_{\mathrm{NIR}}$ and $\mathrm{SR}_{\mathrm{SWIR2.1}}$ represent the surface reflectance of the NIR and SWIR (2.1 μm) bands, respectively. Low-quality pixels, such as those affected by clouds, shadows, aerosols or snow, were masked out using MOD09A1's quality assurance (QA) bits. Additionally, the Collection 6 MODIS MOD44W water mask (Carroll et al. 2016, 2017) was resampled and applied to exclude water bodies. The dNBR was then computed for each Julian Day (JD) using the following formula:

$$\mathrm{dNBR} = \mathrm{NBR}_{\mathrm{year,JD}} - \mathrm{NBR}_{\mathrm{year+1,JD}}$$

where $\mathrm{NBR}_{\mathrm{year,JD}}$ and $\mathrm{NBR}_{\mathrm{year+1,JD}}$ represent NBR values on the same JD in a given year and the following year, respectively. This approach ensures that dNBR is calculated based on anniversary image pairs, a practice recommended by (Chen et al. 2020, Chen, Fu, et al. 2021).

Next, spring dNBR composites were created specifically for tiles MODIS Vertical Tile #1 (V1) and Tile #2 (V2). This step is essential for accurately identifying late-season burned areas in high-latitude environments, where snow cover may set in while fires are still active, obscuring the land surface required for burned area mapping. These composites were generated using the dNBR data processed in the previous step and were limited to JDs 121–153 to capture the late-season burns from the last year in Arctic regions. The spring composites took the last valid dNBR value within the JDs 121–153 window for each tile in a given year.

The workflow then proceeded to identify potential burned areas for all MODIS Vertical Tiles (V1–V4). The fundamental rationale is that for a pixel to be considered potentially burned, it must exhibit high dNBR values. However, the threshold for determining 'high' dNBR values should be lower for shrublands and grasslands, where vegetation cover is sparse. To achieve this, the MODIS MOD44B Vegetation Continuous Fields (VCF; Collection 6.1; Hansen et al. 2013; DiMiceli et al. 2022) dataset was overlaid with dNBR data for each of the 8-day cycles in each year. A pixel on a given 8-day cycle was considered potentially burned if it met one of the following criteria: it had a tree cover value above 10% and a dNBR value greater than 0.25 or it had a tree cover value below 10% and a dNBR value greater than 0.2 (accounting for the lower dNBR values typically associated with fires in shrublands and grasslands). All potential burned areas identified in this step were then evaluated against MODIS MCD14ML active fire detections (Collection 6.1; Giglio et al. 2016). Prior to this evaluation, the active fire data had been reprocessed which involved clipping active fire points from each year plus the last 3 months of the previous year (to account for late-season fires obscured by snow) within a 5-km buffer of individual MODIS tile shapefiles. The preprocessed active fire data was compared with potential burned areas to exclude those unrelated to fires. Specifically, a potential burned polygon was confirmed as burned if it met both of the following criteria: its size was larger than 300 ha and it overlapped with active fires detected within 64 days (i.e., eight 8-day cycles) prior to the date on which the potential burned area was mapped. These spatial and temporal thresholds, introduced by Loboda et al. (2007), were designed to improve the mapping accuracy of northern wildfires.

For high-latitude tiles (i.e., V1 and V2), where spring composites are created, the same processing steps were applied to identify burned areas based on spring composites from the current year and the following year. For example, to map late-season burns that occurred in 2020, spring composites from 2020 were compared with spring composites from 2021. This approach allowed for the detection of late-season fires and the filling in of areas where burn scars were obscured in late-season MODIS imagery due to snow cover or persistent cloud cover.

Finally, the burned area product for a given year was created by merging all burned areas identified for that year, including those detected using spring composites from the following year. This comprehensive approach ensured that the ABBA product captures late-season fires, providing a complete representation of burned areas in the Arctic region.

The assessment of ABBA consisted of two main components: a direct intercomparison of burned areas mapped by ABBA and the MODIS MCD64A1 product and a confusion matrix-based accuracy assessment. In the first evaluation, we compared the interannual burned area reported by ABBA and MCD64A1 between 2002 and 2022. Specifically, burned areas were summarised for two major terrestrial biomes, boreal forests and Arctic tundra, as delineated by the WWF World Ecoregions Map (Olson et al. 2001). The second evaluation focused on assessing ABBA's mapping accuracy using confusion matrices and associated accuracy metrics, with a goal to directly compare the performance of ABBA with that of MCD64A1. All burned area mapped by ABBA and MCD64A1 between 2002 and 2022 was aggregated into two separate cumulative burned area maps. Since both products share the same projection (sinusoidal) and spatial resolution (463.313 m, nominally 500 m), no reprojection or resampling was required. These two maps were then merged into a single map with four unique classes, as defined in Table 2.

Random sample points were selected based on the merged map, with the number of points allocated to each class determined by two key rationales. First, the distribution of sample points for the three burned classes (Classes 2–4) was largely proportional to their respective burned area. Second, the proportion of sample points for burned classes was intentionally boosted compared to the unburned class (Class 1) to allow for a clearer examination of

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the differences in performance between ABBA and MCD64A1. This was necessary because only about 10% of the study area was burned during the 2002–2022 period. Following these rationales, 100 random points were allocated to each of the three burned classes (since their burned areas, as mapped by ABBA, were within the same order of magnitude), while 500 points were assigned to the unburned class.

Each random sample point was validated by analysts who were blinded to the mapping process. The analysts used Landsat imagery to determine whether the point was burned or unburned between 2002 and 2022. If a point was found to have burned, the analyst was asked to record the year of the burn based on the available imagery. Points located in cropland or urban areas were excluded from the analysis to ensure the assessment focused solely on wildfires. Additionally, points with insufficient Landsat imagery for confident validation were also excluded.

TABLE 2 | Class values in the merged cumulative burned area map, their definitions and allocated numbers of random sample points.

Class value	Class definition	Number of points allocated
1	Mapped as unburned by both ABBA and MCD64A1	500
2	Mapped as burned by both ABBA and MCD64A1	100
3	Mapped as burned by ABBA but not by MCD64A1	100
4	Mapped as burned by MCD64A1 but not by ABBA	100

These excluded points accounted for a small portion of the total sample size and were not replenished.

Two confusion matrices were generated, one for ABBA and one for MCD64A1, based on the validated sample points. For each product, the four classes were merged into two categories: Burned (Classes 2 and 3 for ABBA; Classes 2 and 4 for MCD64A1) and Unburned (Classes 1 and 4 for ABBA; Classes 1 and 3 for MCD64A1). A burned pixel was considered correctly mapped only if it was identified as burned and the year of fire matched. Overall accuracy, as well as commission and omission errors for the burned and unburned classes, were calculated.

3 | Results

As shown in Figure 2, ABBA mapped over 2 million km^2 of burned area between 2002 and 2022, substantially more than the total burned area captured by MCD64A1. This difference highlights ABBA's ability to detect a greater extent of wildfire activity in high northern latitudes. The confusion matrices and accuracy metrics presented in Table 3 demonstrate that ABBA not only maps more burned area than MCD64A1 but also does so more accurately. ABBA achieves an overall accuracy of 87.2%, higher than MCD64A1's 82.7% and exhibits lower commission and omission errors for both burned and unburned classes. Specifically, ABBA's commission error for burned areas is 15.6%, compared to MCD64A1's 23.1%, indicating that ABBA is less likely to misclassify unburned areas as burned. Similarly, ABBA's omission error for burned areas is 29.6%, lower than MCD64A1's 39.1%, suggesting that ABBA is better at detecting actual burned areas. For unburned areas, ABBA's omission error is 5.6%, compared to MCD64A1's 7.9%, further underscoring its improved performance. These results emphasise ABBA's enhanced capability to accurately map burned area in high northern latitude regions, making it a more reliable tool for wildfire monitoring and analysis in these regions.

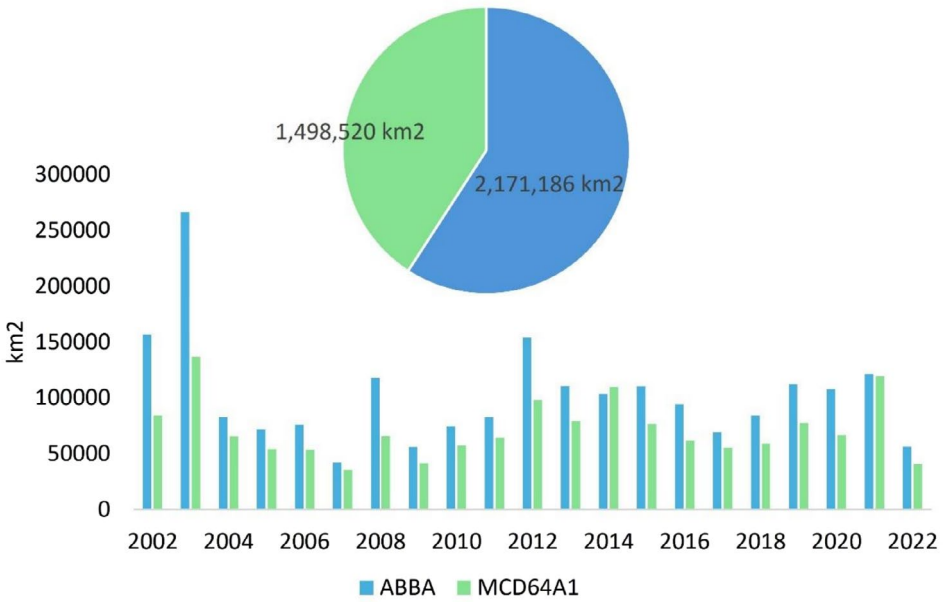

FIGURE 2 | Comparison of interannual and cumulative burned areas mapped by ABBA and MCD64A1.

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TABLE 3 | Confusion matrices and accuracy metrics for ABBA and MCD64A1, generated from the same pool of random sample points. Bold values indicate overall accuracies.

ABBA		**Reference**			
		Burned	Unburned	Total	Commission error
Classified	Burned	162	30	192	15.6%
	Unburned	68	504	572	11.9%
	Total	230	534	764	
	Omission Error	29.6%	5.6%		87.2%
MCD64A1		Reference			
		Burned	Unburned	Total	Commission error
Classified	Burned	140	42	182	23.1%
	Unburned	90	492	582	15.5%
	Total	230	534	764	
	Omission Error	39.1%	7.9%		82.7%

4 | Discussion

ABBA can be utilised in several ways, particularly for advancing our understanding of wildfires in the HNL. As demonstrated in the previous section, ABBA achieves notably higher mapping accuracies than MCD64A1 in these regions. Between 2002 and 2022, ABBA captured over 40% more burned areas, an expanse roughly equivalent to the size of France, highlighting the extensive scope of wildfire activity that MCD64A1 may have missed. This discrepancy suggests that the broader role of HNL wildfires in the Earth's system has likely been underestimated when relying solely on MCD64A1 data.

By providing one of the most accurate representations of HNL wildfire distribution, ABBA can significantly enhance research and policy decisions related to wildfire management and climate change. One of its most critical applications is in the development of more precise wildfire emission products. Currently, estimating fire emissions in HNL regions, particularly across Eurasia, relies heavily on the Global Fire Emissions Database (GFED), which uses MCD64A1 as its primary input for burned area. Consequently, inaccuracies in MCD64A1's data directly affect emission estimates. Given ABBA's broader coverage of burned area, an emission database incorporating ABBA could provide a more comprehensive account of HNL fire emissions, revealing higher levels of carbon and pollutants released into the atmosphere.

However, further work is needed before ABBA can fully replace or supplement MCD64A1 in fire emission inventories. One key limitation is that ABBA, in its current form, does not include date-of-burn (DoB) information, which is typically required for accurate wildfire emission modelling. Emission estimates often depend on the timing of fires to model plume dispersal and assess impacts on atmospheric chemistry. Fortunately, DoB information can be obtained by integrating active fire detections from NASA's Moderate Resolution Imaging Spectroradiometer (MODIS) and the Visible Infrared Imaging Radiometer Suite (VIIRS). While both sensors are useful for monitoring wildfires, VIIRS is generally preferable for future-oriented products due to its higher spatial resolution and longer expected operational lifespan. Given the imminent decommissioning of MODIS instruments, there is a strong need to adapt the current MODIS-based ABBA algorithm to work with VIIRS inputs. This adaptation will ensure the continuation of ABBA's mission to map the long-term spatio-temporal dynamics of wildfires in the HNL.

Beyond wildfire emissions, ABBA's precise spatio-temporal data on HNL wildfires offers considerable benefits for studying various climate and ecological impacts. Many wildfire-driven processes, such as changes in vegetation cover, soil chemistry and water cycles, depend on the extent of burned area. With accurate data on where and how much area is affected, researchers can better assess how wildfires influence broader environmental changes. For instance, improved burned-area maps enable more accurate estimates of shifts in carbon storage, rates of permafrost thaw and vegetation regrowth patterns in boreal and tundra ecosystems.

In terms of climate feedbacks, albedo changes represent one of the most critical factors linking wildfires to global and regional climate dynamics (Randerson et al. 2006; Rogers et al. 2015). Albedo, the fraction of incoming sunlight reflected by a surface, can be significantly altered after a wildfire. The replacement of vegetation by charred surfaces and subsequent regrowth can shift the energy balance, contributing to either warming or cooling effects depending on local conditions. Accurate burned area data is essential for studying these impacts, as post-fire albedo changes are typically calculated on a per-square-meter basis. If burned area estimates are underestimated, spatially integrated measures of climate forcing will also be underestimated, skewing the results of climate models that aim to account for fire-driven feedback loops.

Overall, ABBA stands out as a valuable tool for advancing our understanding of wildfires in high northern latitudes. Its ability to map larger burned areas more accurately than MCD64A1 not

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only improves emission estimates but also deepens our knowledge of wildfire impacts on the climate system. As researchers incorporate date-of-burn information and continue refining ABBA's data products, it is poised to become an indispensable resource for scientists, policymakers and land managers. By capturing a fuller scope of fire activity in these critical regions, ABBA ultimately helps ensure that global models better reflect the real-world influence of HNL wildfires on both the carbon cycle and climate.

4.1 | Product Availability

ABBA was released via the Oak Ridge National Laboratory Distributed Active Archive Centre (ORNL DAAC) at https://doi.org/10.3334/ORNLDAAC/2328. The random sample points used for the validation can be accessed at https://doi.org/10.5281/zenodo.18066065.

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Data Availability Statement

The data that support the findings of this study are openly available in ORNL Distributed Active Archive Center at https://doi.org/10.3334/ORNLDAAC/2328.

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